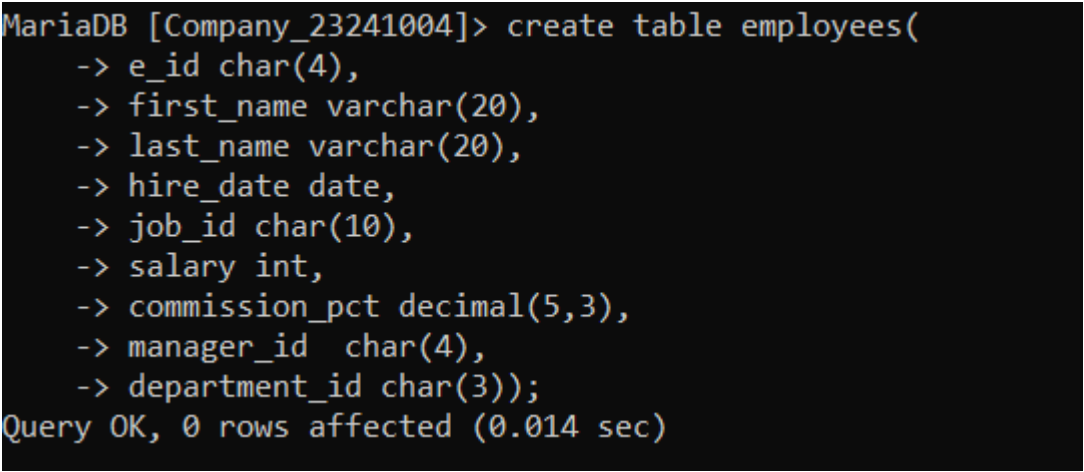


CSE370 : Database Systems

Assignment 02 | Spring 2026

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No 1 Query (as Plain Text)	create table employees(e_id char(4), first_name varchar(20), last_name varchar(20), hire_date date, job_id char(10), salary int, commission_pct decimal(5,3), manager_id char(4), department_id char(3));
No 1 SS (of Query & Output in Shell)	 <pre>MariaDB [Company_23241004]> create table employees(-> e_id char(4), -> first_name varchar(20), -> last_name varchar(20), -> hire_date date, -> job_id char(10), -> salary int, -> commission_pct decimal(5,3), -> manager_id char(4), -> department_id char(3)); Query OK, 0 rows affected (0.014 sec)</pre>
No 2 Query (as Plain Text)	insert into employees values ('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'),

	('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
--	---

No 2 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> insert into employees values -> ('E001','John','Doe','2020-01-15','IT_PROG',65000, NULL,'E010','D01'), -> ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), -> ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), -> ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), -> ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), -> ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'), -> ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), -> ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), -> ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), -> ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), -> ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), -> ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), -> ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), -> ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), -> ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04'); Query OK, 15 rows affected (0.010 sec) Records: 15 Duplicates: 0 Warnings: 0 </pre>
No 3 Query (as Plain Text)	<pre> select first_name, last_name, hire_date, department_id from employees where hire_date=(select max(hire_date) from employees); </pre>

No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> select first_name, last_name, hire_date, department_id -> from employees -> where hire_date=(select max(hire_date) from employees); +-----+-----+-----+-----+ first_name last_name hire_date department_id +-----+-----+-----+-----+ Kevin Lewis 2022-03-12 D04 +-----+-----+-----+-----+ 1 row in set (0.001 sec) </pre>
No 4 Query (as Plain Text)	<pre> select first_name, last_name, e_id, salary, department_id from employees e1 where salary=(select max(salary) from employees e2 where e1. department_id=e2. department_id); </pre>

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23241004]> select first_name, last_name, e_id, salary, department_id  
-> from employees e1  
-> where salary=(select max(salary)  
-> from employees e2  
-> where e1. department_id=e2. department_id  
-> );
```

```
+-----+-----+-----+-----+-----+  
| first_name | last_name | e_id | salary | department_id |  
+-----+-----+-----+-----+-----+  
| Patricia   | Garcia    | E010 | 85000  | D01            |  
| Thomas     | Lee       | E011 | 80000  | D02            |  
| Maria      | White     | E012 | 75000  | D03            |  
| Daniel     | Harris    | E013 | 78000  | D04            |  
+-----+-----+-----+-----+-----+  
4 rows in set (0.001 sec)
```

No 5 Query
(as Plain Text)

```
select first_name, last_name, e_id, commission_pct, department_id  
from employees  
where department_id='D02'and commission_pct> all(select commission_pct  
from employees  
where department_id='D01' and commission_pct is not null);
```

No 5 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> select first_name, last_name, e_id, commission_pct, department_id -> from employees -> where department_id='D02'and commission_pct> all(select commission_pct -> from employees -> where department_id='D01' and commission_pct is not null); +-----+-----+-----+-----+-----+ first_name last_name e_id commission_pct department_id +-----+-----+-----+-----+-----+ Jane Smith E002 0.150 D02 Emily Davis E004 0.200 D02 James Taylor E007 0.180 D02 Thomas Lee E011 0.100 D02 +-----+-----+-----+-----+-----+ 4 rows in set (0.002 sec) </pre>
No 6 Query (as Plain Text)	<pre> select department_id, count(*) from employees group by department_id having max(salary) <=70000; </pre>
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> select department_id, count(*) -> from employees -> group by department_id -> having max(salary) <=70000; Empty set (0.001 sec) </pre>
No 7 Query	<pre> select department_id, job_id, salary </pre>

(as Plain Text)	from employees e1 where salary < any(select salary from employees e2 where e1.department_id= e2. department_id);
No 7 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> select department_id, job_id, salary -> from employees e1 -> where salary < any(select salary -> from employees e2 -> where e1.department_id= e2. department_id); +-----+-----+-----+ department_id job_id salary +-----+-----+-----+ D01 IT_PROG 65000 D02 SA_REP 55000 D01 IT_PROG 62000 D02 SA_REP 58000 D01 IT_PROG 60000 D03 HR_REP 50000 D02 SA_REP 60000 D04 FI_ACC 52000 D01 IT_PROG 67000 D03 HR_REP 48000 D04 FI_ACC 51000 +-----+-----+-----+ 11 rows in set (0.001 sec) </pre>
No 8 Query (as Plain Text)	select distinct manager_id from employees e1

	where manager_id is not null and not exists (select * from employees e2 where e2.manager_id = e1.manager_id and e2.salary<55000);
No 8 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23241004]> select distinct manager_id -> from employees e1 -> where manager_id is not null -> and not exists (select * -> from employees e2 -> where e2.manager_id = e1.manager_id -> and e2.salary<55000); +-----+ manager_id +-----+ E010 E011 +-----+ 2 rows in set (0.001 sec) </pre>
No 9 Query (as Plain Text)	select first_name, last_name, e_id, salary, department_id, commission_pct from employees e1 where commission_pct = (select max(commission_pct) from employees e2 where e1.manager_id=e2.manager_id);

No 9 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23241004]> select first_name, last_name, e_id, salary, department_id, commission_pct
-> from employees e1
-> where commission_pct = (select max(commission_pct)
-> from employees e2
-> where e1.manager_id=e2.manager_id);
```

first_name	last_name	e_id	salary	department_id	commission_pct
Emily	Davis	E004	58000	D02	0.200
Sarah	Brown	E006	50000	D03	0.150

```
2 rows in set (0.002 sec)
```

No 10 Query
(as Plain Text)

```
select job_id, avg(salary)
from employees
group by job_id
having avg(salary)< (select avg(salary)
from employees);
```

No 10 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23241004]> select job_id, avg(salary)
-> from employees
-> group by job_id
-> having avg(salary)< (select avg(salary)
-> from employees);

+-----+-----+
| job_id | avg(salary) |
+-----+-----+
| FI_ACC | 51500.0000 |
| HR_REP | 49000.0000 |
| SA_REP | 57666.6667 |
+-----+-----+
3 rows in set (0.001 sec)
```